

PERN Platform

Complete Technical Documentation

*Pollution & Environmental Risk Navigator
Full-Stack IoT Environmental Intelligence Platform*

PostgreSQL . Express . React . Node.js
Version 2.7.0

*Complete technical overview of the PERN Platform, including system architecture,
features, scientific foundations, AI core, device connectivity, and virtual sensor system.*

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9. Virtual Sensor Estimation System

The Virtual Sensor Estimation System estimates missing sensor values from available physical measurements using scientifically-validated models. The system is organized into 4 tiers based on what physical sensors are available, starting from just Temperature + Humidity (DHT22, ~\$5) and scaling up to full environmental monitoring with MQ-135 gas sensor, light sensor, and water sensors.

Each estimator includes a "Real sensor" field showing what dedicated physical sensor would measure that value directly. This helps users understand the tradeoff between estimated accuracy and hardware cost.

9.1 Tier 1: Temperature + Humidity (4 estimates)

Requires only temperature and humidity sensors (e.g., DHT22, BME280, SHT30). Highest confidence.

Dew Point (C)

$$T_d = 237.7 * G / (17.27 - G), \quad G = 17.27 * T / (237.7 + T) + \ln(RH/100)$$

R-squared:	0.97
Confidence:	94%
Inputs:	Temp 55%, Humidity 45%
Real sensor:	Chilled mirror hygrometer / capacitive dew-point sensor

Heat Index (C)

$$\text{NOAA Rothfusz 9-term polynomial: } HI = -42.379 + 2.049 * T_f + \dots$$

R-squared:	0.95
Confidence:	92%
Inputs:	Temp 55%, Humidity 45%
Real sensor:	Composite index derived from temp + humidity

Vapor Pressure Deficit (kPa)

$$VPD = e_s - e_a, \quad e_s = 0.6108 * e^{(17.27 * T / (T + 237.3))}$$

R-squared:	0.93
Confidence:	90%
Inputs:	Temp 55%, Humidity 45%
Real sensor:	Composite index derived from vapour pressure + temp

WBGT (C)

$$WBGT = 0.7 * T_w + 0.3 * T, \quad T_w \text{ via Stull (2011)}$$

R-squared:	0.92
Confidence:	88%
Inputs:	Temp 55%, Humidity 45%
Real sensor:	WBGT meter (ISO 7243) / heat stress monitor

9.2 Tier 2: + MQ-135 Gas Sensor (5 estimates)

Add an MQ-135 gas sensor (~\$3) for air quality estimates. The MQ-135 is sensitive to NOx, CO, NH3, smoke.

NO2 (ppb)

$$NO_2 = MQ135 * 120 + (T - 25) * 0.8 + (50 - RH) * 0.3$$

R-squared:	0.65
Confidence:	62%
Inputs:	MQ135 55%, Temp 25%, Humidity 20%
Real sensor:	Electrochemical NO2 sensor (e.g., NO2-B43F)

CO2 (ppm)

$$CO_2 = 400 + MQ135 * 400 + (T - 24) * 18 + (RH - 65) * 3$$

R-squared:	0.68
Confidence:	65%
Inputs:	MQ135 50%, Temp 30%, Humidity 20%
Real sensor:	NDIR CO2 sensor (e.g., MH-Z19B, SCD30)

PM2.5 (ug/m3)

$$PM2.5 = MQ135 * 65 + \max(0, RH - 60) * 0.3$$

R-squared:	0.58
Confidence:	55%
Inputs:	MQ135 60%, Humidity 25%, Temp 15%
Real sensor:	Laser particle counter (e.g., PMS5003)

AQI (est.) (0-500)

$$AQI = MQ135 * 180 + \text{cold inversion penalty}$$

R-squared:	0.62
Confidence:	60%
Inputs:	MQ135 60%, Temp 25%, Humidity 15%
Real sensor:	Composite AQI (EPA method, multiple pollutants)

Pollution Density ()

$$EPD = (Gas/1000) * (1 + T/50) + RH/200$$

R-squared:	0.76
Confidence:	74%
Inputs:	MQ135 45%, Temp 30%, Humidity 25%
Real sensor:	Composite pollution index (no single sensor)

9.3 Tier 3: + Light Sensor (4 estimates)

Add a light sensor (LDR ~\$1, TSL2561 ~\$3, VEML7700 ~\$4) for solar and plant-growth metrics.

UV Index (0-15)

$$\text{Lux piecewise: } <2K=0, 2-10K=0-3, 10-30K=3-7, 30-60K=7-11, >60K=11+$$

R-squared:	0.74
Confidence:	70%
Inputs:	Light 70%, Temp 30%
Real sensor:	UV photodiode (e.g., VEML6075, GUVA-S12SD)

Solar Radiation (W/m2)

$$\text{Solar} = \text{Lux} / 119 \text{ (luminous efficacy 119 lm/W)}$$

R-squared:	0.88
Confidence:	85%
Inputs:	Light 75%, Temp 25%
Real sensor:	Pyranometer (e.g., Apogee SP-110)

PPFD (umol/m2/s)

$$PPFD = \text{Lux} * 0.0185 \text{ (sunlight conversion at 550nm)}$$

R-squared:	0.80
Confidence:	76%
Inputs:	Light 80%, Temp 20%
Real sensor:	PAR sensor (e.g., Apogee SQ-420)

Evapotranspiration (mm/day)

$$ET = 0.0023 * R_s * (T + 17.8) * \sqrt{T} / 10$$

R-squared:	0.85
Confidence:	80%
Inputs:	Light 50%, Temp 35%, Humidity 15%
Real sensor:	Composite index (FAO56, no single sensor)

9.4 Tier 4: Water Sensors (4 estimates)

When water sensors (pH, TDS, Water Temp, DO) are present, aquatic estimates are generated.

Dissolved Oxygen (mg/L)

$$DO_{\text{sat}} = 14.6 - 0.4T + 0.008T^2 - 0.0001T^3 \text{ (APHA 4500-O)}$$

R-squared:	0.89
Confidence:	82%

Inputs: Water Temp 55%, pH 25%, TDS 20%

Real sensor: Electrochemical/optical DO sensor (Atlas Scientific, SEN0237)

pH (0-14)

$7.0 + (DO-7)*0.15 + (25-T)*0.008 + \text{TDS buffer}$

R-squared: 0.72

Confidence: 68%

Inputs: DO 40%, Water Temp 35%, TDS 25%

Real sensor: pH probe (e.g., Atlas Scientific pH, SEN0161)

NH3 (ppm)

$NH3 = 1/(1+10^{(pKa-pH)})*20 + DO \text{ stress penalty}$

R-squared: 0.82

Confidence: 76%

Inputs: pH 50%, Water Temp 30%, DO 20%

Real sensor: Electrochemical NH3 sensor (e.g., MQ-137)

Turbidity (NTU)

$NTU = TDS/30 + \max(0, 7-DO)*2.5 + |pH-7|*1.5$

R-squared: 0.70

Confidence: 66%

Inputs: TDS 40%, DO 30%, pH 15%, Water Temp 15%

Real sensor: Nephelometric turbidity sensor (e.g., SEN0189)

9.5 Confidence Scoring

Each virtual sensor includes a multi-factor confidence score (20-96%) calculated from: Input coverage (40%) - ratio of available vs. maximum inputs; Weight quality (25%) - average input weight; Model base accuracy (20%) - expert-assigned quality; R-squared (15%) - scientific validation fit. Scores above 70% = High, 50-70% = Medium, below 50% = Low. Categories: Excellent (green), Good (cyan), Moderate (amber), Poor (orange), Critical (rose).

9.6 UI & Coverage Visualization

The Virtual Sensors page at /virtual-sensors provides a comprehensive view: Stats bar (physical count, estimated count, avg confidence, high-confidence count); Coverage bar (4 tier indicators with unlock/lock status and estimate counts); Physical sensors section (green-tinted cards); Estimated sensors section (tier-grouped, cyan tint, with confidence bars, formulas, input breakdowns, citations); Real sensor info on each estimated card showing what dedicated hardware would measure that value; Locked tier preview (greyed-out cards showing what sensors to add to unlock more estimates).

8. References

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